COER UNIVERSITY

7th km, Roorkee-Haridwar Highway, Haridwar, Uttarakhand

Lab Sheet-09: Mini Projects on Data Analytics and Visualization

MCA III Semester (Session 2026–2027)

Project 1: COVID-19 Data Analysis Dashboard

Objective: Analyze COVID-19 datasets to visualize confirmed cases, recoveries, deaths, vaccination trends, and country-wise statistics.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Simulated COVID-19 Dataset Creation
data = {
    'Country': ['India', 'USA', 'Brazil', 'UK', 'France', 'Germany', 'Russia', 'Tu
rkey'],
    'Confirmed': [45000000, 100000000, 38000000, 24000000, 39000000, 38000000, 230
00000, 17000000],
    'Recovered': [44000000, 98000000, 36000000, 23500000, 38000000, 37500000, 2200
0000, 16800000],
    'Deaths': [530000, 1120000, 700000, 220000, 167000, 174000, 400000, 100000],
    'Vaccinated_Doses': [2200000000, 670000000, 480000000, 150000000, 154000000, 1
92000000, 180000000, 150000000]
}
df = pd.DataFrame(data)

# 1. Exploratory Data Analysis (EDA)
df['Death_Rate (%)'] = (df['Deaths'] / df['Confirmed']) * 100
df['Recovery_Rate (%)'] = (df['Recovered'] / df['Confirmed']) * 100

print("--- Dataset Overview ---")
print(df.describe())

# 2. Visualizations
plt.figure(figsize=(14, 10))

# Chart 1: Confirmed Cases vs Deaths
plt.subplot(2, 2, 1)
sns.barplot(x='Confirmed', y='Country', data=df, palette='viridis')
plt.title('Total Confirmed Cases')
plt.xlabel('Cases (in 100 Millions)')

# Chart 2: Death Rates
plt.subplot(2, 2, 2)
```

```
sns.barplot(x='Death_Rate (%)', y='Country', data=df, palette='Reds_r')
plt.title('Death Rate Comparison (%)')

# Chart 3: Vaccination Trends
plt.subplot(2, 2, 3)
plt.pie(df['Vaccinated_Doses'], labels=df['Country'], autopct='%1.1f%%', startangl
e=140)
plt.title('Vaccination Doses Distribution')

# Chart 4: Correlation Heatmap
plt.subplot(2, 2, 4)
sns.heatmap(df[['Confirmed', 'Recovered', 'Deaths', 'Vaccinated_Doses']].corr(), a
nnot=True, cmap='coolwarm')
plt.title('Metric Correlations')

plt.tight_layout()
plt.savefig('covid_dashboard.png')
plt.show()
```

Project 2: Sales Data Analysis Dashboard

Objective: Develop an interactive sales analysis dashboard showing monthly sales, profit, customer segmentation, product performance, and regional analysis.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Generate Mock E-commerce Sales Data
np.random.seed(42)
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'N
ov', 'Dec']
regions = ['North', 'South', 'East', 'West']
segments = ['Consumer', 'Corporate', 'Home Office']

data_rows = []
for month in months:
    for region in regions:
        for segment in segments:
            sales = np.random.randint(10000, 50000)
            profit = sales * np.random.uniform(0.1, 0.3)
            data_rows.append([month, region, segment, sales, profit])

sales_df = pd.DataFrame(data_rows, columns=['Month', 'Region', 'Segment', 'Sales',
 'Profit'])

# Data Aggregation & Analysis
monthly_summary = sales_df.groupby('Month')[['Sales', 'Profit']].sum().reindex(mon
ths)
regional_summary = sales_df.groupby('Region')['Sales'].sum().reset_index()
segment_summary = sales_df.groupby('Segment')['Profit'].sum().reset_index()

# Visualizations Dashboard
fig, axes = plt.subplots(2, 2, figsize=(15, 10))
```

```
# Chart 1: Monthly Sales & Profit Trends
axes[0, 0].plot(monthly_summary.index, monthly_summary['Sales'], marker='o', color
='b', label='Sales')
axes[0, 0].plot(monthly_summary.index, monthly_summary['Profit'], marker='s', colo
r='g', label='Profit')
axes[0, 0].set_title('Monthly Sales and Profit Performance')
axes[0, 0].legend()

# Chart 2: Regional Sales Breakdown
sns.barplot(x='Region', y='Sales', data=regional_summary, ax=axes[0, 1], palette='
Blues_r')
axes[0, 1].set_title('Regional Sales Distribution')

# Chart 3: Customer Segment Analysis
axes[1, 0].pie(segment_summary['Profit'], labels=segment_summary['Segment'], autop
ct='%1.1f%%', colors=['#ff9999','#66b3ff','#99ff99'])
axes[1, 0].set_title('Profit Contribution by Segment')

# Chart 4: Sales vs Profit Scatter
sns.scatterplot(x='Sales', y='Profit', hue='Region', data=sales_df, ax=axes[1, 1])
axes[1, 1].set_title('Sales vs Profit Breakdown')

plt.tight_layout()
plt.savefig('sales_dashboard.png')
plt.show()
```